Revision

Revision Index	Datum des Eintrags	Änderungs-beschreibung	Verant-wortlicher, Name, Unterschrift, Datum	CR Nummer	Bemerkung	Seite
1	2	3	4	5	6	7
-	02.08.2021	New drawing	D. Ehrlich	--	-	0
a	14.10.2021	Update: starter wiring/relays KF1.0a&b/relais wiring	D. Ehrlich /M. Mayer	CR020513	CN029547	0

A Rolls-Royce
solution

designation

circuit-diagram "control-panel" BR4000ECU9 DEIF AGC-4 A10 VDE4110

project number

XZ54934000032 (ZNG00040841)

material number

see table-item

serial number

type-of-system

SUP42

			creator	E56395	02.08.2021	customer-description	project-description	A Rolls-Royce solution	Page Description	drawing-number	project number	EPLAN P8	==	
			editor.	Dieter Ehrlich	21.10.2021		circuit-diagram "control-panel" BR4000ECU9 DEIF AGC-4 A10 VDE4110	Rolls-Royce Solutions GmbH	Cover sheet	XZ54934000032		2.8.3	+	&EAA =
			checked	F.Jäger	10.03.2020					material number	status	format	version	page (&EAA) 1 from 10
revision	date	name	norm	M.Prinz	10.03.2020					see table-item	with Connectivity	A3	Standard	Full pages 2 from 92

PROJEKT INFORMATION

Wire labelling:	No	
Protection:	IP54	
Mains connection:	Rated voltage	400V AC
	Mains frequency:	50Hz
	Wires of cable:	3 Phase / N / PE
	Cross section max.:	5x10mm²
	Max load:	22A
	Max Icu:	25kA

Part list number	remark
X54934400441	Deif VDE4110 Grid Code VDE4110 6A
X54934400442	Deif VDE4110 Grid Code VDE4110 6A mit IOT Device
X54934400443	Deif VDE4110 Grid Code VDE4110 6B
X54934400444	Deif VDE4110 Grid Code VDE4110 6B mit IOT Device

INFORMATION TO SWITCHGEAR ASSEMBLY

	generally	auxiliary-drives	Generator	voltage sensing busbar / mains
type-of-electrical-system	TN-S			
rated operating-voltage Ue:		max. 440 VAC	max. 690 VAC	max. 690 VAC
rated voltage Un:		440 VAC	690 VAC	690 VAC
rated insulation voltage Ui:		440 VAC	690 VAC	690 VAC
rated impulse withstand voltage Uimp		4 kV	6 kV	6 kV
rated current In:		**3,0 bis 25,0A**		
max. back-up-fuse:				
rated conditional short-circuit current Icc:				
rated diversity factor RDF:	**0,08 - 0,68**			
rated frequency fn:	50 Hz			
overvoltage-category:	III			
pollution-degree:	3			
EMC ambience:	A			
protection-class:	IP 54			
min. ambient-temperature:	0 °C			
max. ambient-temperature:	45 °C			
max. installation-altitude:	2000 m			
auxiliary-voltage DC:	24 V			
auxiliary-voltage AC:	24 V			

Type label:

Client:	# # #
Series no.:	# # #
Type designation:	MIP BR4000
Material no.:	# # #
Type of electricyl system:	TN-S
Rated Voltage Un:	415V AC
Rated Frequency fn:	50 Hz
Rated current InA:	# # #
Rated short current ICC:	25 kA
Control Voltage:	24V DC
Protection class:	IP 54
Year of manufacturing:	# # #
Listed standards:	IEC 61439-1 IEC 61439-2 DIN EN 60204-1

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			editor.	Dieter Ehrlich	21.10.2021		circuit-diagram "control-panel" BR4000ECU9 DEIF AGC-4 A1 0 VDE4110
			checked	F.Jäger	10.03.2020		
revision	date	name	norm	M.Prinz	10.03.2020		

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TECHNISCHE DATEN UND KONSTRUKTIONS DETAILS

IEC 61355

Vollständige Bezeichnung	Strukturbeschreibung
mounting-location	
+BSF	Grundrahmen
+ENG	Motor
+GEN	Generator inkl. Klemmenkasten
+MIP	Motor-Steuer-Schrank
+MMC	MTU Modul Steuerschrank
+GCB	Generator Leistungsschalter
+MCB	Netz Leistungsschalter
+FAN	Lüfter Steuerschrank
+TNK	Kraftstoff-Tank
+EXT	Nicht im Schaltschrank eingebaute Geräte und Gehäuse die nicht von MTU geliefert werden
+CUS	
+DOK	Dokumente

Vollständige Bezeichnung	Strukturbeschreibung
document-type	
&EAA	Deckblatt
&EAB	Inhaltsverzeichnis
&EDB	Beschreibung
&EFS	Schaltplan
&ETL	Layout
&EMA	Klemmenplan
&EMB	Kabelplan
&EPC	Materialstückliste

Vollständige Bezeichnung	Strukturbeschreibung
system	
=EKW	Elektrische Stromversorgung
=CON	Steuerspannung
=EES	Maschinen Not-Halt
=PLC	SPS oder Genset Kontroller
=COM	Kommunikation
=STR	Motor Start
=MCC	Gemisch Kühl-Kreis
=ECC	Motor/Kühlwasser System
=HWC	Heiz-Wasser-Kreis
=FUL	Kraftstoff Versorgung
=MSC	Sonstiges

TECHNICAL DATA AND CONSTRUCTION DETAILS

line and color definition for potential tracing

(Colors are only visible if you use colored print out)

Supply, Line and measure voltage	L1, L2, L3 (120V, 230V, 400V ,690V, ...)	L
Supply, Line and measure voltage	N (120V, 230V, 400V ,690V, ...)	N
Protective earth	Ground	PE
Control voltage + 24VDC, 12VDC L+	L+	L+
Control voltage - 0VDC L-	L-	L-
Shield	SH	shield
Starter battery cabeling +24V	L-	L+Batt
Starter battery cabeling -24V	L-	L-Batt

line and color definition for potential tracing

(Colors are only visible if you use colored print out)

Control voltage 24VAC 24VAC L		24VAC-L
Control voltage 24VAC 24VAC N		24VAC-N
External source voltage	....	ext
Analog signals	QSOLL+, QSOLL-	analog
Current transformer	L1-l-Geno, L1-k-Geno	L1-k-Geno
Emergency stop	L+	NA
All other connections EPLAN standard		

TECHNICAL DATA AND CONSTRUCTION DETAILS

Wire colors within enclosure

According
IEC EN 60204-1

Rated voltage	Above 48VAC	black
Neutral wire	N	light blue
Protective earth	PE	green/yellow
Control voltage	24-240V AC	red
	0V AC	red/white
	24-110V DC	dark blue
	0V DC	grey
PLC-Controller		white
Measuring technique		grey
wires before Mainswitch		orange

Main power before main switch	L1 - L2 - L3
Main power main switch on	1L1 - 1L2 - 1L3
Neutral wire	N
Protective conductor	PE

Mains power supply	400VAC	-XD00
Power supply Heating GEN	230VAC	-XD01
Power supply heatg. cool. water	400VAC	-XD02
Emergency stop		-XD06
Supply DC	24VDC	-XDL+
Supply DC	0VDC	-XDL-
Harness W010-ECU7 / W011-ECU9 Digital		-1XD01
Harness W010-ECU7 / W011-ECU9 Analog		-1XD02
Harness W020 Engine		-2XD01
Harness W033 Generator signal		-3XD01 /-3XD02 / -3XD03
Option		
Customer interface WD04.1 / WG04.2 /		-4XD01 / -4XD02 /
WG04.3 / WD04.4		-4XD03 / -4XD04 /
WG04.5		-4XD05
Harness W050 Power panel		-5XD01
Control cable WD06.1 Generator circuit breaker		-6XD01
Control cable WD07.1 Mains circuit breaker		-7XD01
Control cable WD08.1 Fan control enclosure		-8XD01
Control cable WD09.1/WD09.2 Fuel control		-9XD01/-9XD02
CAN bus		-XF01

Tightening torque

Power relay	75A	M3,5x5 M5x8	1,1-1,2 Nm 3,2-3,5 Nm
Starter	Prestolite 9KW	M5 M12	2,0-3,0 Nm 27-33 Nm
Starter	Prestolite 15KW	M5 M10	1,5-1,7 Nm 9-10 Nm
Starter	Delco 9KW	Housing 10-32 INCH 1/2-13 INCH	18-23 Nm 1,8-3,4 Nm 27-34 Nm
Battery disconnect switches		M10 M12	15-20 Nm 18-22 Nm
Battery terminals	DIN 72331	Cable M6 M8	5-6 Nm Class 4,6 / 3,5 Nm Class 4,6 / 8,4 Nm
Grounding		M8 M10 M12 M14	Class 8,8 / 28 Nm Class 8,8 / 54 Nm Class 8,8 / 93 Nm Class 8,8 / 148 Nm